AI-Powered Smart Water Reservoir Management and Demand Forecasting Platform

COURSE NAME: SOFTWARE ENGINEERING
COURSE CODE: CSA1013

Submitted by:
VISHAL B
192524485
B.TECH AI&DS

PROBLEM STATEMENT:

Software Requirement Specification (SRS)

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.
Total Maximum Marks	**20 Marks**			

1. Introduction

Purpose

This Software Requirement Specification (SRS) document defines the functional and non-functional requirements for the AI-Powered Smart Water Reservoir Management and Demand Forecasting Platform. It is intended to guide the design, development, and validation of a system that monitors reservoir conditions, forecasts water demand, predicts water availability, and supports data-driven distribution and drought-planning decisions.

Business Domain and Stakeholders

The system operates in the water resource management domain, serving government/municipal water authorities, reservoir operations teams, agricultural stakeholders, and industrial consumers of water. Key stakeholders include reservoir operators, water resource managers, farmers/irrigation officers, system administrators, and regulatory authorities responsible for sustainable water policy.

1.3 Existing Challenges

Traditional reservoir management relies heavily on manual observation and historical averages, which limits the ability to anticipate droughts, heavy rainfall events, or sudden demand shifts. This often results in inefficient allocation, water shortages, or wastage, and delayed response to climate variability.

System Objectives

1. Monitor reservoir water levels in real time.
2. Forecast water demand across drinking, irrigation, and industrial sectors.
3. Predict future water availability using AI/ML models.
4. Optimize water distribution across competing demands.
5. Analyze rainfall and weather patterns affecting inflow.
6. Generate water management and regulatory reports.
7. Support proactive drought planning and early warning.
8. Improve overall water resource utilization and sustainability.

2. Problem Description

Water reservoirs are critical infrastructure for supplying drinking water, irrigation, and industrial needs. Climate variability, population growth, and reliance on manual, historically-driven decision-making create persistent challenges in balancing water availability against demand. During droughts, insufficient early warning leads to shortages; during heavy rainfall, poor coordination can lead to overflow risk or wasted inflow. An AI-powered reservoir management platform addresses this gap by continuously monitoring reservoir levels, rainfall forecasts, inflow/outflow rates, and demand patterns, and by using predictive analytics to recommend optimal, sustainable distribution strategies.

3. Scope of the System

The system will cover end-to-end reservoir intelligence: data acquisition from IoT/SCADA sensors and external weather/satellite sources; AI-based forecasting of demand and availability; optimization and recommendation of distribution strategies; alerting for drought/flood risk; and reporting for internal and regulatory stakeholders. The initial scope targets a pilot deployment across a defined set of reservoirs with role-based web and mobile access for operators, managers, field officers, and administrators. Out of scope for this phase: direct automated control (actuation) of physical dam gates/valves, and billing/payment processing for water usage.

4. Functional Requirements

The following functional requirements define the core capabilities the system must provide.

ID	Requirement	Description
FR-1	Reservoir Level Monitoring	The system shall continuously collect and display real-time reservoir water level, storage volume, and capacity percentage data from IoT/SCADA sensors.
FR-2	Inflow/Outflow Tracking	The system shall record and display inflow and outflow rates at reservoir intake and release points, updated at configurable intervals.
FR-3	Rainfall & Weather Analysis	The system shall ingest rainfall forecasts and satellite/weather data and analyze historical and predicted rainfall patterns for the catchment area.
FR-4	Water Demand Forecasting	The system shall use historical consumption data and AI/ML models to forecast short-term (daily/weekly) and long-term (seasonal) water demand for drinking, irrigation, and industrial use.
FR-5	Water Availability Prediction	The system shall predict future water availability by combining current storage, forecasted inflow, rainfall predictions, and demand forecasts.
FR-6	Distribution Optimization	The system shall recommend optimal water distribution/release strategies across sectors (drinking, irrigation, industrial) based on predicted availability and demand.
FR-7	Drought & Flood Alerting	The system shall generate automated alerts and early warnings when reservoir levels fall below drought thresholds or rise above flood-risk thresholds.
FR-8	Report Generation	The system shall generate periodic (daily/weekly/monthly) water management reports summarizing levels, demand, allocation, and forecasts, exportable as PDF/Excel.
FR-9	Dashboard & Visualization	The system shall provide role-based dashboards with charts, maps, and trend graphs for reservoir status, forecasts, and allocation plans.
FR-10	Sensor/Device Management	The system shall allow administrators to register, configure, calibrate, and monitor the health of IoT sensors and data-feed integrations.
FR-11	User Authentication & Access Control	The system shall authenticate users and enforce role-based access control (Operator, Manager, Farmer/Officer, Administrator).
FR-12	Threshold & Rule Configuration	The system shall allow authorized users to configure alert thresholds, forecasting parameters, and allocation priority rules.
FR-13	Historical Data Archival	The system shall archive historical reservoir, weather, and demand data to support trend analysis and model retraining.
FR-14	Notification Delivery	The system shall deliver alerts and reports to stakeholders via SMS, email, and in-app notifications.

5. Non-Functional Requirements

The following non-functional requirements define the quality attributes the system must satisfy.

Category	Requirement Description
Performance	The system shall process real-time sensor data and refresh dashboards within 5 seconds, and generate demand forecasts within 2 minutes for a 30-day horizon.
Scalability	The system shall support monitoring of at least 100 reservoirs and 10,000 IoT sensor endpoints concurrently, scaling horizontally via cloud infrastructure.
Reliability	The system shall maintain 99.5% uptime and provide failover data collection (local caching) during connectivity loss, with data sync on reconnection.
Security	The system shall encrypt data in transit (TLS 1.2+) and at rest (AES-256), enforce role-based access control, and maintain audit logs of all critical actions.
Usability	The system shall provide an intuitive, map-based, multilingual web and mobile interface usable by non-technical field operators with minimal training.
Maintainability	The system shall use a modular, API-driven microservices architecture to allow independent updates to forecasting models, sensor integrations, and UI.
Availability	The system's monitoring and alerting functions shall be available 24/7, including during scheduled maintenance windows via redundant deployment.
Interoperability	The system shall integrate with third-party weather/satellite APIs, government hydrology databases, and SCADA systems via standard protocols (REST, MQTT).
Data Accuracy	AI demand and availability forecasts shall maintain a minimum accuracy (e.g., MAPE $\leq$ 10%) validated against historical actuals, with periodic model retraining.
Portability	The system shall be deployable across major cloud providers (AWS/Azure/GCP) and support containerized deployment (Docker/Kubernetes).

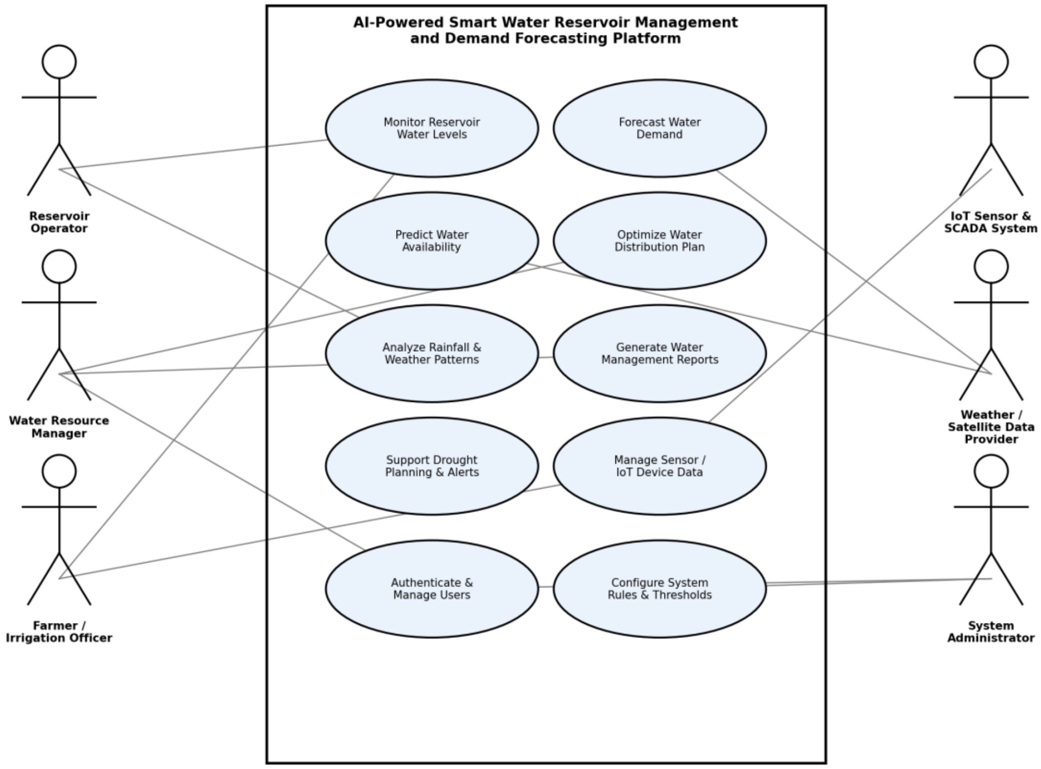
6. Actors and Use Cases

Identified Actors

Actor	Type	Description
Reservoir Operator	Primary	Monitors real-time reservoir levels, inflow/outflow, and sensor health at the field/site level; responds to alerts.
Water Resource Manager	Primary	Reviews forecasts and availability predictions, approves distribution strategies, and oversees drought planning at the organizational level.
Farmer / Irrigation Officer	Primary	Views water allocation schedules and demand forecasts relevant to irrigation, and submits demand requests.
System Administrator	Primary	Manages user accounts, roles, sensor/device configuration, and system-wide thresholds and rules.
IoT Sensor & SCADA System	Secondary (System)	Automatically supplies real-time water level, flow, and quality data to the platform.
Weather / Satellite Data Provider	Secondary (System)	Supplies rainfall forecasts, satellite imagery, and meteorological data via external APIs.
Municipal / Government Water Authority	Secondary	Receives consolidated reports for regional water policy and drought-preparedness decisions.

Use Case Diagram

The diagram below illustrates the interaction between the identified actors and the major use cases of the AI-Powered Smart Water Reservoir Management and Demand Forecasting Platform.



Use Case Descriptions

UC-1: Monitor Reservoir Water Levels

Use Case ID	UC-1
Use Case Name	Monitor Reservoir Water Levels
Actor(s)	Reservoir Operator, IoT Sensor & SCADA System
Description	Allows the operator to view real-time and historical reservoir water level, storage volume, and capacity data collected from sensors.
Pre-condition	Sensors are installed, calibrated, and transmitting data; operator is authenticated.
Main Flow	1. Sensor system streams water level and flow data to the platform. 2. System validates and stores incoming data. 3. Operator opens the monitoring dashboard. 4. System displays current level, trend graph, and capacity percentage. 5. System raises an alert if levels cross configured thresholds.
Post-condition	Reservoir status is up to date and visible on the dashboard; alerts are logged if thresholds are breached.

UC-2: Forecast Water Demand

Use Case ID	UC-2
Use Case Name	Forecast Water Demand
Actor(s)	Water Resource Manager
Description	Enables the manager to generate AI-based demand forecasts for drinking, irrigation, and industrial sectors over a selected time horizon.
Pre-condition	Historical consumption data and current sector inputs are available.
Main Flow	1. Manager selects a reservoir and forecast horizon (e.g., 7/30/90 days). 2. System retrieves historical demand and seasonal patterns. 3. AI/ML model generates a sector-wise demand forecast. 4. System displays the forecast with confidence intervals. 5. Manager exports or shares the forecast report.
Post-condition	A demand forecast is generated and available for planning and reporting.

UC-3: Predict Water Availability & Optimize Distribution

Use Case ID	UC-3
Use Case Name	Predict Water Availability & Optimize Distribution
Actor(s)	Water Resource Manager
Description	Combines forecasted inflow, rainfall predictions, and demand forecasts to predict future water availability and recommend an optimal distribution plan.
Pre-condition	Demand forecast (UC-2) and rainfall analysis (UC-5) have been completed.
Main Flow	1. Manager requests an availability prediction for a defined period. 2. System aggregates current storage, forecasted inflow, and rainfall data. 3. System runs the optimization model to balance supply against sector demand. 4. System proposes a distribution plan with allocation percentages per sector. 5. Manager reviews and approves or adjusts the plan.
Post-condition	An approved water distribution plan is recorded and made available to downstream stakeholders.

UC-4: Support Drought Planning & Alerts

Use Case ID	UC-4
Use Case Name	Support Drought Planning & Alerts
Actor(s)	Water Resource Manager, System (automated)
Description	Automatically evaluates reservoir and forecast data against drought risk thresholds and notifies stakeholders.
Pre-condition	Drought thresholds and notification rules are configured.

Main Flow	1. System continuously evaluates storage trend against configured drought thresholds. 2. System detects a projected shortage condition. 3. System generates a drought risk alert and recommended mitigation actions. 4. Notification is sent to the manager and relevant authorities. 5. Manager reviews recommendations and initiates a drought response plan.
Post-condition	Stakeholders are alerted to drought risk in advance and mitigation actions are documented.

UC-5: Manage Sensor / IoT Device Data

Use Case ID	UC-5
Use Case Name	Manage Sensor / IoT Device Data
Actor(s)	System Administrator, IoT Sensor & SCADA System
Description	Allows the administrator to register, configure, and monitor the health/status of connected sensors and data feeds.
Pre-condition	Administrator is authenticated with appropriate privileges.
Main Flow	1. Administrator adds/configures a new sensor or data source endpoint. 2. System validates connectivity and begins data ingestion. 3. System continuously monitors sensor health (signal, battery, data gaps). 4. System flags malfunctioning or offline sensors. 5. Administrator schedules maintenance or replacement as needed.
Post-condition	Sensor inventory and health status are current; data quality is maintained.

7. Data Requirements

Data Entity	Description
Reservoir Master Data	Reservoir ID, name, location (GPS), maximum capacity, catchment area, associated sectors served.
Sensor Telemetry Data	Timestamp, sensor ID, water level, inflow rate, outflow rate, water quality parameters (optional).
Weather & Rainfall Data	Forecasted/actual rainfall, temperature, humidity, satellite-derived soil moisture, source and timestamp.
Demand & Consumption Data	Historical and current consumption by sector (drinking/irrigation/industrial), region, and time period.
Forecast & Prediction Data	Generated demand forecasts, availability predictions, confidence intervals, model version used.
Distribution Plan Data	Approved allocation percentages/volumes per sector, effective period, approver, approval timestamp.
Alert & Notification Data	Alert type, severity, triggering condition, timestamp, recipients, acknowledgement status.
User & Role Data	User ID, name, role, assigned reservoirs/regions, contact details, authentication credentials (hashed).

8. External Interface Requirements

Interface Type	Description
User Interface	Responsive web dashboard and mobile app with role-based views, interactive maps, charts, and report export (PDF/Excel).
Hardware Interface	IoT water-level, flow, and quality sensors connected via MQTT/LoRaWAN/cellular gateways to the ingestion service.
Software / API Interface	REST APIs for weather/satellite data providers, government hydrology databases, and SCADA system integration.
Communication Interface	HTTPS/TLS for web and API traffic; SMS and email gateways for alert notifications; push notifications for mobile app.

9. System Features

Feature	Description
Real-Time Monitoring Dashboard	Consolidated, map-based view of all reservoirs showing current levels, trends, and active alerts.
AI Demand Forecasting Engine	Machine-learning module producing short- and long-term sector-wise water demand forecasts.
Water Availability & Optimization Engine	Predicts future availability and recommends optimal distribution across competing demands.

Feature	Description
Alerting & Early Warning Module	Rule- and ML-driven detection of drought/flood risk with multi-channel notifications.
Reporting Module	Automated, schedulable generation of management and regulatory reports.
Sensor & Device Management Module	Registration, calibration tracking, and health monitoring of connected IoT devices.
Administration & Access Control Module	User, role, and configuration management with full audit trail.

10. Assumptions and Constraints

Assumptions

1. Reservoirs are equipped, or will be equipped, with IoT-enabled sensors capable of transmitting data over a network.
2. Reliable internet or cellular connectivity is available at reservoir sites, or local data caching is used to bridge outages.
3. Historical water level, rainfall, and demand data of sufficient quality and duration are available to train forecasting models.
4. Users (operators, managers, officers) have basic digital literacy to operate the web/mobile dashboard.
5. Third-party weather and satellite data providers offer stable, documented APIs with acceptable licensing terms.
6. Government/municipal authorities will cooperate in supplying regional consumption and policy data where required.

Constraints

Constraint Type	Description
Technical	The system depends on the accuracy and availability of external weather/satellite APIs and the reliability of installed IoT hardware.
Operational	Field sensor installation and calibration require trained personnel and periodic on-site maintenance visits.
Economic	Budget limits the number of reservoirs that can be instrumented and the choice of cloud infrastructure tier in initial phases.
Legal / Regulatory	The system must comply with regional water-resource governance regulations and data-sharing agreements with government bodies.
Time	The project must deliver a functional pilot (limited reservoirs) within the sanctioned project timeline before full-scale rollout.
Data Privacy	Consumption data tied to specific farms/consumers must be handled in compliance with applicable data protection requirements.

11. Requirement Completeness and Consistency Validation

Stakeholder Coverage Check

All identified stakeholders — reservoir operators, water resource managers, farmers/irrigation officers, system administrators, and regulatory authorities — are mapped to at least one actor and associated use case set, confirming that their operational needs (monitoring, forecasting, allocation approval, field-level visibility, system administration, and regulatory reporting) are addressed in the functional requirements.

Ambiguity, Redundancy, and Consistency Check

- No conflicting requirements were identified; alerting (FR-7) and reporting (FR-8) are complementary rather than overlapping, since alerts are real-time/event-driven while reports are periodic/summary-driven.
- Terminology was standardized across the document (e.g., 'water availability prediction' vs 'demand forecasting' are consistently distinguished as separate, dependent capabilities).
- Each functional requirement is traceable to at least one system objective and one use case, avoiding orphan or unjustified requirements.
- Non-functional requirements were cross-checked against functional requirements to ensure feasibility (e.g., the 5-second dashboard refresh in NFR-Performance is achievable given the described sensor data ingestion architecture).

Identified Gaps and Suggested Improvements

- Add an explicit data-retention/archival policy requirement to clarify how long historical sensor and forecast data must be retained for regulatory audits.
- Define a formal model-accuracy monitoring requirement (e.g., automated alerts when forecast error exceeds an acceptable threshold) to keep AI predictions trustworthy over time.
- Clarify offline/degraded-mode behavior further — e.g., maximum acceptable duration of cached/local operation before manual intervention is required.
- Consider adding a requirement for multi-reservoir/basin-level aggregated views for authorities managing several interconnected reservoirs.

Conclusion

The requirement set is judged complete and internally consistent for the pilot scope defined in Section 3. The suggested improvements above are recommended as refinements for subsequent requirement review iterations rather than as blocking gaps.